Choose a motion method

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose a motion relationship and justify its assumptions.

SUCCESS CRITERIA - I can...

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “constant” and “interval” to explain your model or result. Sentence starters are optional.

constant: not changing during the stated interval

interval: the time or range between two stated values

acceleration: rate of change of velocity

01 Bring back an earlier idea

x_i=-3±0.5 m, v=5±0.1 m/s and exact t=7 s.
Give central final position, its conservative interval and full width.
Name an assumption.

Word help: final = at the end of the interval; subscript f

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_f = v_i + a\Delta t$; $\Delta x = v_i\,\Delta t + \frac{1}{2}a(\Delta t)^2$
$v_f^2 = v_i^2 + 2\,a\Delta x$.
Constant acceleration; i = initial, f = final. Choose the equation containing your knowns and unknown.

Worked reasoning: From rest, a cart accelerates uniformly at 2 m/s² for 3 s. Final velocity=6 m/s; average velocity=(0+6)/2=3 m/s; displacement=3×3=9 m.

Your turn: Show the same 9 m using the area of a triangle on a velocity-time graph.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Solve and check

A cart begins at x=-3 m with v_i=2 m/s and constant a=1 m/s² for 4 s.
Find final velocity, displacement and final position.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

04 No time given

A bike moves at 12 m/s and brakes at constant -3 m/s² to rest.
Find stopping distance and then time as a check.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 At the top

A ball is thrown upward at 14 m/s.
Use g=9.8 m/s² and neglect drag.
Find maximum rise and time to the top. Is acceleration zero there?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

06 Choose a method

Velocity changes from -3 to -15 m/s in 3 s, then stays at -15 m/s for 7 s.
Find acceleration and describe the speed trend in each interval.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A car starts at +12 m/s and brakes with a=-3 m/s² until it stops. A student extends the equation to 8 s and reports v=-12 m/s.
Explain what assumption fails and give stopping time/distance.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A cart starts at 2 m/s and travels 24 m in 4 s with uniform acceleration. Solve for acceleration in two ways and explain why a constant-speed method fails.

Word help: uniform = constant throughout the stated interval

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3